StockSync Insights

Retail Product Demand Prediction and Inventory Analytics

Statistics and Data Analytics - Activity 2 mini project | 1 November 2026

Abstract

A reproducible Python pipeline analyzed 1,067,371 real retail transaction rows and predicted weekly units sold for 50 products. The last 12 complete weeks were held out. Random Forest achieved MAE 251.286, RMSE 426.056, and R^2 0.278; a previous-week baseline achieved MAE 258.475. The improvement is modest and does not justify automatic purchasing decisions.

Problem and dataset

The goal is to predict product demand from sales already observed. The source is UCI Online Retail II: two years of invoices from a UK non-store giftware retailer. It records sales, prices, dates and product codes, but not stock-on-hand or product images. These products are unrelated to StockSync's live grocery inventory. The dataset is CC BY 4.0 and is credited below.

Workbook SHA-256: bcbe73b35f5b7babf197fb0cb983a11f5d9ff929078d4aa53d171b1f2df2e980

Cleaning and weekly aggregation

Both Excel sheets were loaded. Cancelled invoices, returns, nonpositive prices, non-UK rows, invalid product codes, missing values, and incomplete boundary weeks were excluded in that order. Valid transaction quantities were grouped into Monday-Sunday product-weeks; absent sales weeks were filled with zero.

Stage	Rows
Source	1,067,371
Cancelled	19,494
Returns / nonpositive quantity	3,457
Nonpositive price	2,750
Outside UK	83,169
Invalid product code	3,772
Complete-week retained sales	924,913

Method and chronological evaluation

The top 50 eligible products were selected using only pretest sales, with at least 20 active sales weeks. A global Random Forest regressor predicts a product-week's units from sales lags 1-4, their mean, last observed unit price, calendar week, and one-hot product identity. All predictors precede their target week. Settings: 100 trees, maximum depth 12, minimum leaf size 3, random seed 42.

Training: 4,281 product-weeks before 12 September 2011. Holdout: 600 product-weeks from 12 September to 4 December 2011. Baseline: predict the prior week's units. This time-ordered split avoids training on later sales when predicting earlier weeks.

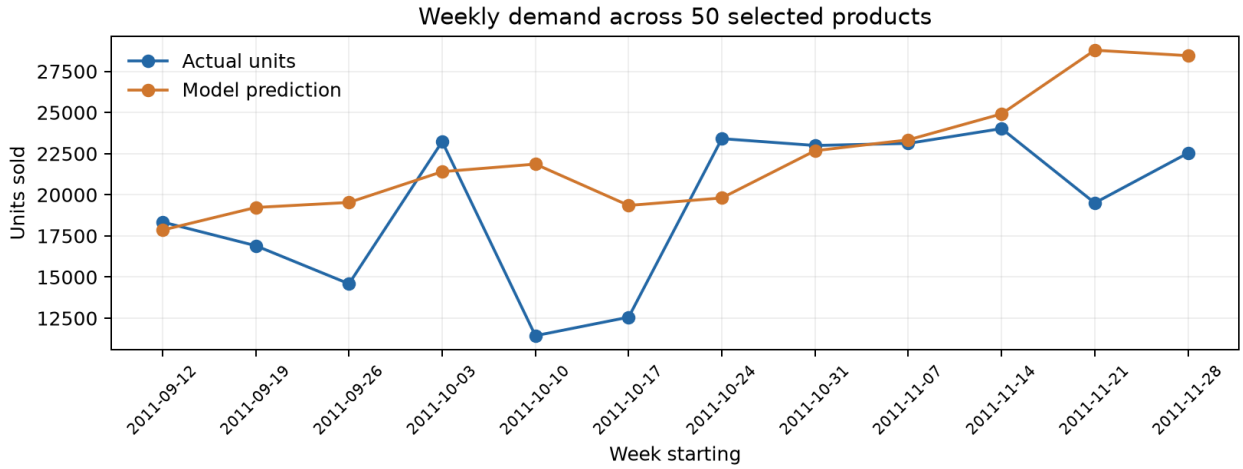


Figure 1. Aggregated actual and predicted units across the 50 test products. Weekly spikes remain difficult to predict.

Measured results

Method	MAE	RMSE	R^2
Random Forest	251.286	426.056	0.278
Previous week	258.475	507.413	-0.024

MAE and RMSE are errors in units per product-week (lower is better). R^2 can be negative; higher is better. The model improves MAE by 7.189 units (2.8%) and RMSE by 81.357 units (16.0%) against the baseline. Large residual errors remain.

Interpretation and application

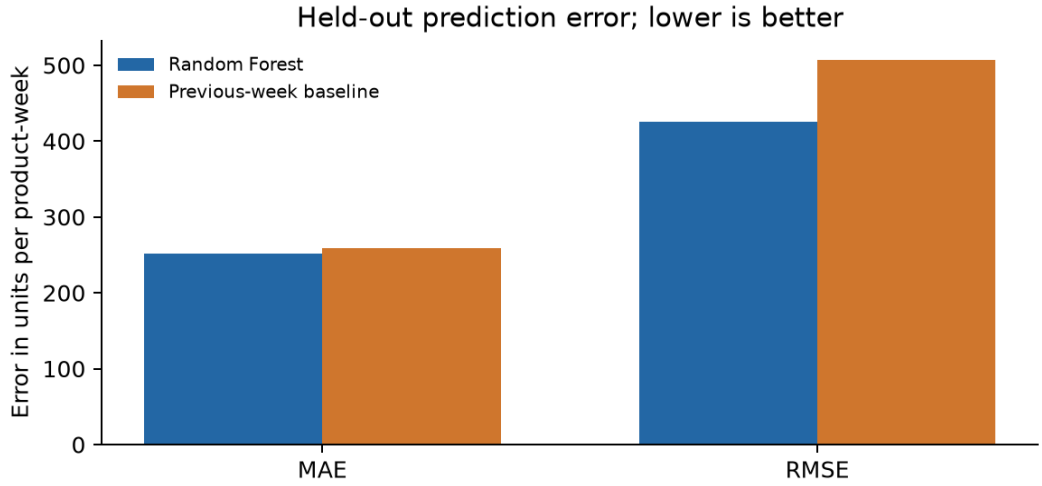


Figure 2. Error comparison on the same 600 held-out product-weeks.

The StockSync Retail Insights page bundles a small, versioned JSON file with the measured metrics and product series. The raw workbook is never uploaded to Firestore. The displayed estimate for the week starting 5 December 2011 is a dated illustration from a model retrained on all complete weeks; it is not an evaluated current forecast or stock recommendation.

Limitations

This UK retailer and its historical giftware and wholesale demand differ from a current Indian grocery store. Prices are GBP, while StockSync uses INR. Transactions show observed sales, not unmet demand or inventory levels. Promotions, stockouts, holidays, and supplier delays are not represented. One 12-week test period is not evidence of stable performance across stores or seasons.

Reproduction

Download UCI Online Retail II, verify the workbook SHA-256 on page 1, install analytics/requirements.txt, and run: python analytics/build.py --input /path/to/online_retail_II.xlsx. Run python -m unittest discover -s analytics -p 'test_*.py', then python analytics/render_report.py. The generated JSON records source, cleaning, model settings, holdout, metrics, and per-product predictions.

Sources

Chen, D. (2012). Online Retail II. UCI Machine Learning Repository. DOI: 10.24432/C5CG6D.

Scikit-learn: Lagged features for time series forecasting and time-ordered evaluation.